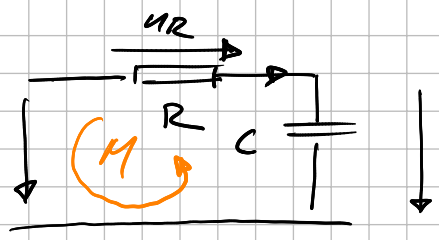
Ersatzschaltbild

System Eingang $u = u_E$



$u_C = y$ System Ausgang

1 DGL

$i_R = i_C = i$

$u_R = R i$

$i_C = C \dot{u}_C$

Maschensatz

$u_E = \underbrace{u_R}_{RC\dot{u}_C} + u_C$

$\underbrace{RC\dot{u}_C}_{i_C}$

$$\boxed{\dot{x}(t) \triangleq \frac{dx(t)}{dt}}$$

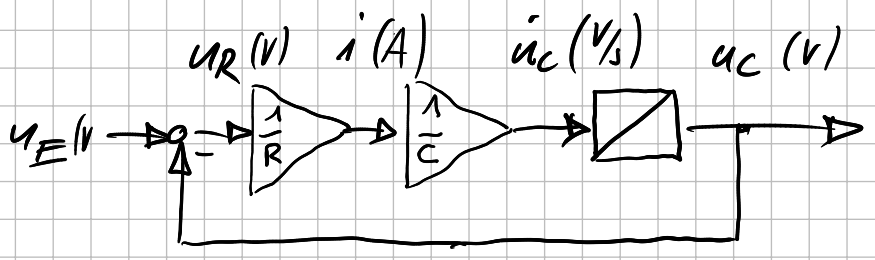
$\rightarrow RC\dot{u}_C + u_C = u_E$ //

$\left(RC \, d/dt \, u_C(t) + u_C(t) = u_E(t) \right)$

2 Blockschaltbild

$RC\dot{u}_C + u_C = u_E \rightarrow \dot{u}_C = \frac{1}{RC}(u_E - u_C)$

1ne Ableitung → DGL 1ster Ordnung → 1 Integrator



$u_R = u_E - u_C$

$i = \frac{1}{R} u_R$

$i_C = i$

$\dot{u}_C = \frac{1}{C} i_C$

Sprungantwort

$$h(t) = \left(1 - e^{-\frac{t}{RC}}\right)$$

Impulsantwort

$$g(t) = \frac{d}{dt} h(t) = \frac{1}{RC} e^{-\frac{t}{RC}}$$

3 Kennwerte der Sprungantwort

Allgemeine Form

$$T\dot{y} + y = K u$$

K: statische Verstärkung

T: Zeitkonstante

$K = 1\ V/V$ //

$T = RC$ sec //

$$h(t) = K\left(1 - e^{-\frac{t}{T}}\right)$$

$$h(T) = K\left(1 - e^{-1}\right) = K \cdot 0{,}6321$$

$$h(2T) = K\left(1 - e^{-2}\right) = K \cdot 0{,}8647$$

$$h(3T) = \vdots = K \cdot 0{,}9502$$

$$h(4T) = \vdots = K \cdot 0{,}9817$$

$$h(5T) = \vdots = K \cdot 0{,}9933$$

$$h(6T) = \vdots = K \cdot 0{,}9975$$

$$\vdots \qquad \vdots$$

$$h(nT) = K\left(1 - e^{-n}\right)$$

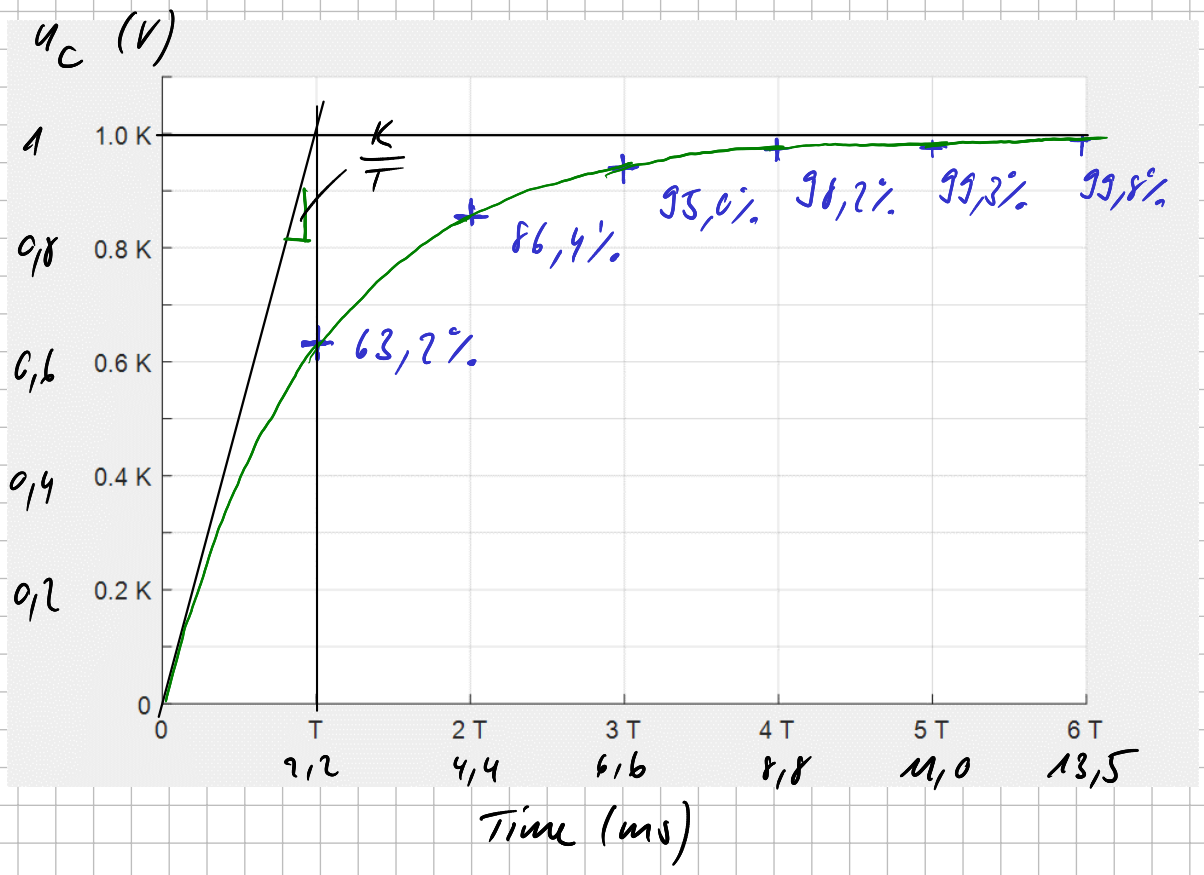
4 Steigung der Sprungantwort bei $t = 0$ sec

$$h(t) = K\left(1 - e^{-\frac{t}{T}}\right)$$

Steigung bei $t = 0$

$$\dot{h}(t) = \frac{K}{T} e^{-\frac{t}{T}} \Big|_{t=0} = \frac{K}{T} \;//$$

5 Skizze Sprungantwort



$$T = RC = 2{,}209 \text{ msec} \;//$$

Frequenzgang $\longleftrightarrow$ Übertragungsfunktion

$\mathcal{F}$	DGL	$\mathcal{L}$
t		t
x $\quad X(\omega)$	$j\omega = s$	x $\quad X(s)$
$\dot{x}$ $\quad (j\omega)\,X(\omega)$	$\longleftrightarrow$	$\dot{x}$ $\quad s\,X(s)$
$\vdots$		$\vdots$
x^n $\quad (j\omega)^n X(\omega)$		x^n $\quad s^n X(s)$

Beispiel

$$G(\omega) = \frac{b_0}{(j\omega)^2 + a_1 (j\omega) + a_0} \qquad j\omega = s \qquad G(s) = \frac{b_0}{s^2 + a_1 s + a_0}$$

6 Übertragungsfunktion $\quad (j\omega \to s)$

$$G(s) = \frac{1}{\underbrace{RC}_{T}\, s + 1} = \frac{B(s)}{A(s)}$$

- Zeitkonstante $T = RC = 2{,}209$ msec //
- statische Verstärkung $K = G(0) = 1$ V/V //
- Nullstellen $B(s) \overset{!}{=} 0 \to z = \{\}$ //
- Polstellen

$$A(s) \overset{!}{=} 0$$

$$RCs + 1 \overset{!}{=} 0 \to p = -\frac{1}{RC} \left(= -\frac{1}{T}\right) //$$

$$= -452{,}7 //$$

7 Frequenzgang

$$RC\underbrace{(j\omega)\,U_C(\omega)}_{\mathcal{F}\{\dot{u}_C\}} + \underbrace{U_C(\omega)}_{\mathcal{F}\{u_C\}} = \underbrace{U_E(\omega)}_{\mathcal{F}\{u_E\}}$$

$$(RCj\omega + 1)\,U_C(\omega) = U_E(\omega)$$

$$\rightarrow \quad U_C(\omega) = \underbrace{\frac{1}{RCj\omega + 1}}_{G(\omega)} U_E(\omega) \quad //$$

$$\boxed{G(\omega) = G(s)\Big|_{s=j\omega}}$$

Amplituden + Phasengang

$$|G(\omega)| = \frac{1}{\sqrt{1+(RC\omega)^2}} \qquad \angle G(\omega) = -\operatorname{atan}(RC\omega)$$

8

$$\boxed{dB(x) = 20\log_{10}(x)}$$

$$\boxed{\omega_g = -p = \frac{1}{RC} = \frac{1}{T}} = 452{,}7 \ \mathrm{rad/sec} \ //$$

$$|G(\omega_g)| = \frac{1}{\sqrt{2}}\,\frac{V}{V} = 0{,}7071\,\frac{V}{V} \approx 0{,}7\,\frac{V}{V} \ //$$

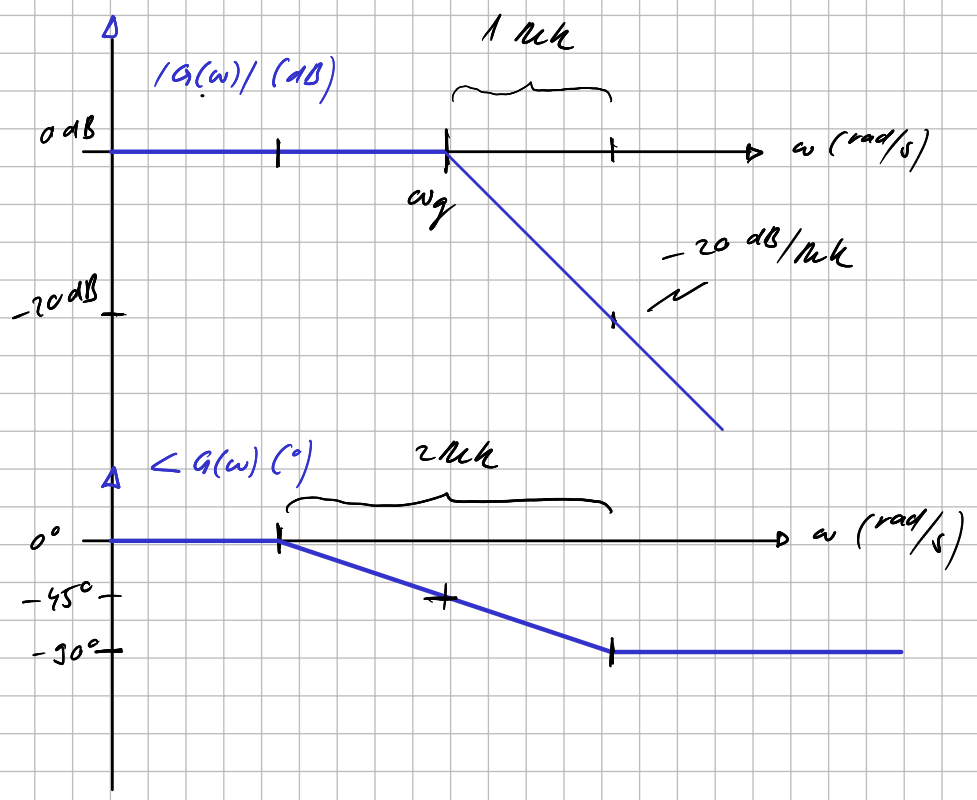
$$|G(\omega_g)|_{dB} = -3{,}0103\ dB \approx -3\,dB \ //$$

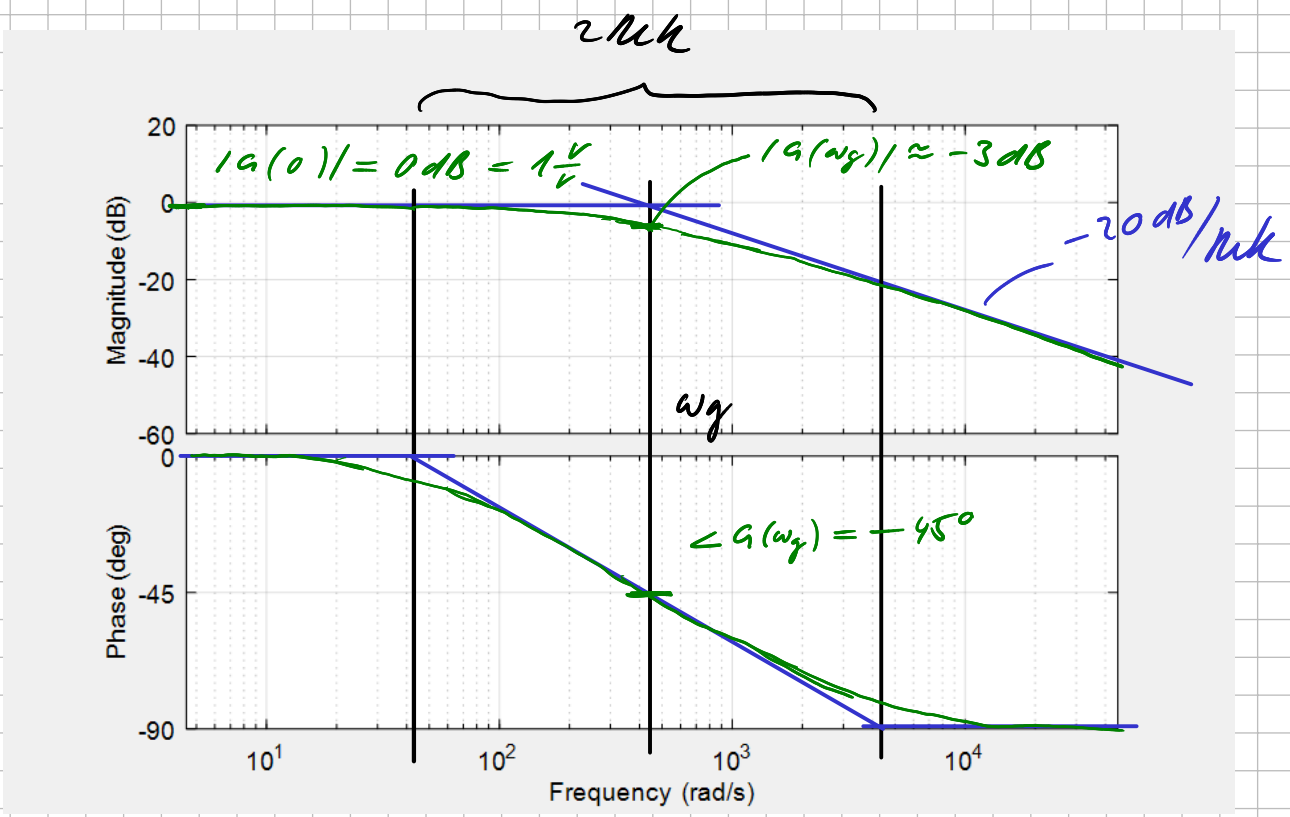
$$\angle G(\omega) = -\operatorname{atan}(1) = -\frac{\pi}{4} \ //$$

$$\angle G(\omega)^\circ = -45^\circ \ //$$

g

Skizze


|G(ω)| (dB)
1 Dek
0 dB
ω (rad/s)
ωg
-20 dB/Dek
-20 dB
∠G(ω) (°)
2 Dek
0°
ω (rad/s)
-45°
-90°

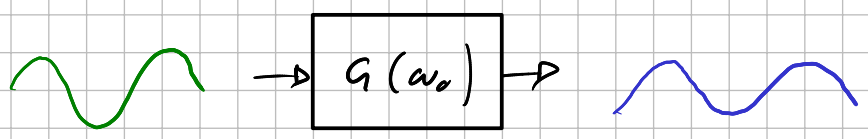

2 Dek
|G(0)| = 0 dB = 1 V/V
|G(ωg)| ≈ -3 dB
-20 dB/Dek
ωg
∠G(ωg) = -45°
Magnitude (dB)
Phase (deg)
Frequency (rad/s)

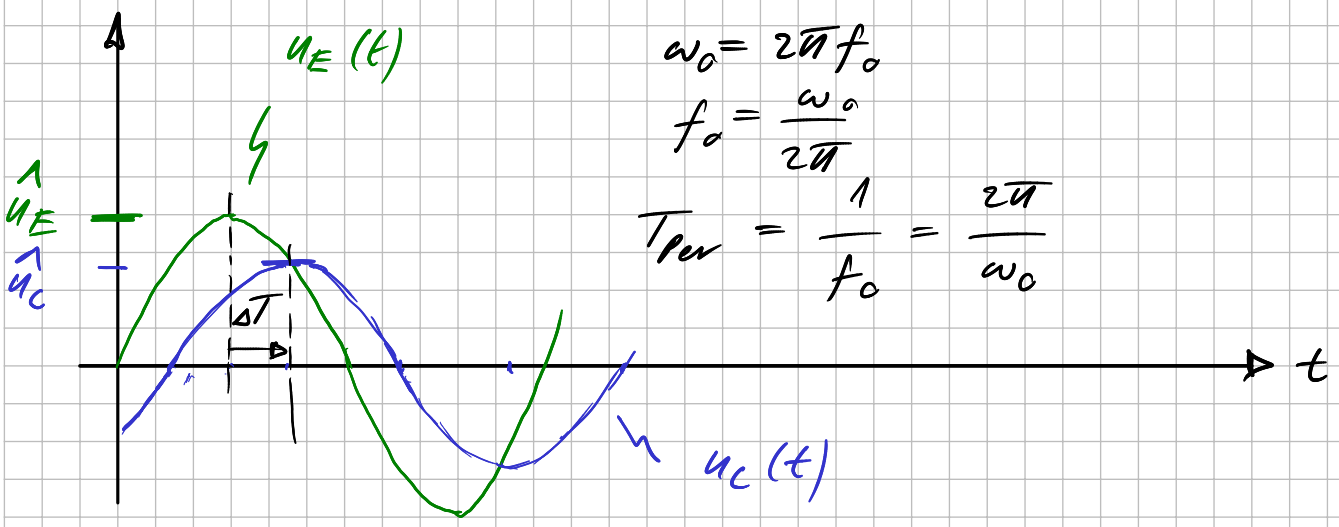
10 Bestimmung des Amplitudenganges und des Phasenganges für eine Frequenz ω_0

$$u_E(t) = \hat{u}_E \sin(\omega_0 t)$$

$$u_C(t) = \underbrace{\hat{u}_C}_{\hat{u}_E |G(\omega_0)|} \sin(\omega_0 t + \angle G(\omega_0))$$

$$G(\omega_0) = |G(\omega_0)| \, e^{j \angle G(\omega_0)}$$





$$\omega_0 = 2\pi f_0$$

$$f_0 = \frac{\omega_0}{2\pi}$$

$$T_{per} = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$$

Amplitudengang

$$\hat{u}_C = |G(\omega_0)| \, \hat{u}_E$$

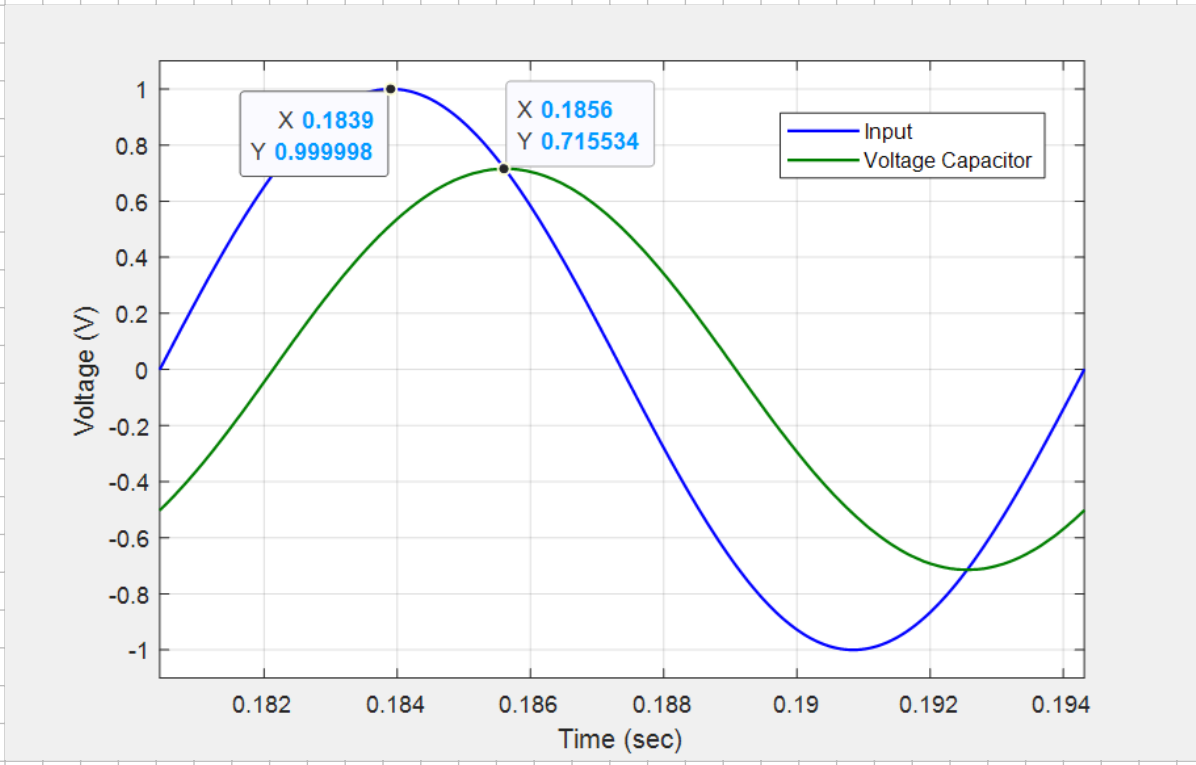
$$\boxed{|G(\omega_0)| = \frac{\hat{u}_C}{\hat{u}_E}}$$

Phasengang

$$T_{per} \triangleq 2\pi$$

$$-\Delta T \triangleq \angle G(\omega_0)$$

$$\boxed{\angle G(\omega_0) = -\frac{\Delta T}{\underbrace{T_{per}}_{\frac{2\pi}{\omega_0}}} 2\pi = -\Delta T \, \omega_0}$$



$$\omega_0 = \omega_g$$

$$|G(\omega_g)| = \frac{\hat{u}_C}{\hat{u}_E} = \frac{0{,}716}{1{,}000}\,\frac{V}{V} = 0{,}716\,\frac{V}{V} \;//$$

$$\angle G(\omega_g) = -\Delta T\,\omega_g = -(0{,}1856\,\text{sec} - 0{,}1839\,\text{sec})\cdot 452{,}7\,\frac{\text{rad}}{\text{sec}}$$

$$= -0{,}770\ \text{rad} = -44{,}1^\circ \;//$$